

NexusIQ Corporation — Inventory Shortage Root Cause Analysis

November 2024 Electronics Stockout Post-Mortem | Reference: OPS-RCA-2024-011

Prepared by: Supply Chain Analytics | Date: December 10, 2024 | Classification: Internal

1. Background

This root cause analysis examines the inventory shortage that developed across NexusIQ's Electronics category during November 2024. The shortage was first identified when routine inventory monitoring flagged 31 Electronics SKUs below reorder point on December 1, 2024 — up from 12 on November 1. By December 15, the network-wide below-reorder count had grown to 72 SKUs across all categories, with Electronics accounting for 39 of those positions.

This analysis should be read in conjunction with the Seasonal Demand Incident Report (OPS-2024-Q4-001), which covers the demand-side events. This document focuses specifically on the supply chain process failures and the inventory management decisions that allowed shortage conditions to develop.

2. Shortage Scope by Product

The shortage was concentrated in Laptop and Tablet SKUs, consistent with those products' higher velocity and longer vendor lead times. Phone shortages developed later in November as the initial Laptop demand drew customers toward Phone as an alternative gift purchase, accelerating Phone depletion unexpectedly.

Product	SKUs Affected	Stores Below Safety Stock	Stockout Days (est.)	Units Short (est.)	Revenue Impact (est.)
Laptop	8 SKU variants	16 stores	4.2 days avg	1,420 units	~\$1,480,000
Tablet	6 SKU variants	11 stores	3.1 days avg	840 units	~\$620,000
Phone	5 SKU variants	8 stores	1.8 days avg	380 units	~\$340,000
Headphones	3 SKU variants	6 stores	1.2 days avg	290 units	~\$87,000
Total Electronics	22 SKU variants	26+ unique stores	2.6 days avg	2,930 units	~\$2,527,000

Estimates based on demand signal analysis at comparable stores that maintained stock. Stockout days measured from first reported zero-inventory date to emergency restock arrival.

3. Root Cause: Systematic Reorder Point Underestimation

The primary root cause was that Electronics reorder points were calibrated on FY 2023 demand data without adjustment for FY 2024 demand growth. Electronics unit velocity grew 18.3% year-over-year from FY 2023 to FY 2024, driven by expanded customer base and improved Electronics category conversion rates. However, the reorder point formula in the inventory management system used a rolling 12-month average that at the time of Q4 2024 planning (September 2024) still included the lower-demand months of Q4 2023 and Q1 2024.

A reorder point calculation using only the most recent 6 months (April-September 2024) would have set the Laptop reorder point at 1,180 units versus the actual 970 units — a 21.6% increase. At the actual Q4 Laptop velocity, the higher reorder point would have triggered purchase orders 8.4 days earlier in the cycle, providing adequate buffer for the Black Friday demand spike.

4. Contributing Factor: Vendor Delivery Variance

TechSource Global's 94.2% on-time delivery rate in FY 2024 means that approximately 5.8% of Electronics orders arrived later than the contracted lead time. During a normal operating period, this variance is absorbed by safety stock without customer impact. During Q4's high-demand window, the same absolute delivery variance had significantly larger consequences because there was no safety stock buffer remaining to absorb delays.

Specifically, two TechSource shipments scheduled to arrive November 20-22 (before Black Friday) arrived November 27-28 (during Black Friday). The combined late inventory of 847 Laptop and Tablet units arrived after the demand peak had already drawn down store stock to critical levels. Had those shipments arrived on schedule, the shortage at the four highest-impact stores would likely have been prevented.

5. Corrective Actions and Implementation Status

Corrective Action	Owner	Target Date	Status	Expected Impact
Update reorder formula to 6-month rolling supply for high-velocity SKUs	Supply Chain Analytics	Jan 15, 2025	In Progress	Earlier reorder triggers; reduces stockout risk by estimating demand more accurately
Add 1.3x seasonal multiplier to Electronics category quantities (in addition to 1.2x multiplier)	Category Ops	Completed Mar 2024	Complete	Larger Q4 safety buffer
Qualify secondary Electronics vendor; target 15% share	Procurement	Mar 31, 2025	In Progress	Reduces single-vendor dependency
Set TechSource on 90-day performance watch; escalate if below 95% on-time	Procurement	Jan 1, 2025	Active	Pressure vendor to improve delivery reliability
Implement promotional demand review for bundles	Category Management	Feb 1, 2025	Planned	Prevents bundle-driven demand surprise
Add automated 30-day Q4 advance order triggers for top 20 Electronics SKUs	Supply Chain	Q2 2025	Planned	Pre-positions stock before demand materializes

6. Lessons Learned

This shortage demonstrated that static reorder parameters calibrated annually cannot keep pace with a rapidly growing business. NexusIQ's Electronics category grew faster than any prior planning cycle anticipated, and the inventory system's backward-looking calibration created a structural lag. The corrective actions above address the immediate process gaps, but the deeper lesson is that inventory parameters must be reviewed on a rolling quarterly basis — not just at annual planning — for high-velocity, high-ASP categories like Electronics.

A secondary lesson is that promotional planning and supply chain planning must be connected earlier in the promotional calendar. The Black Friday Laptop bundle was finalized in marketing in late October without a supply chain impact assessment; future promotions of this scale will require supply chain sign-off no later than six weeks before the promotional start date.